

Project: 2

Data Classification using AI

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Course: Artificial Intelligence Internship

Organization: Decode labs



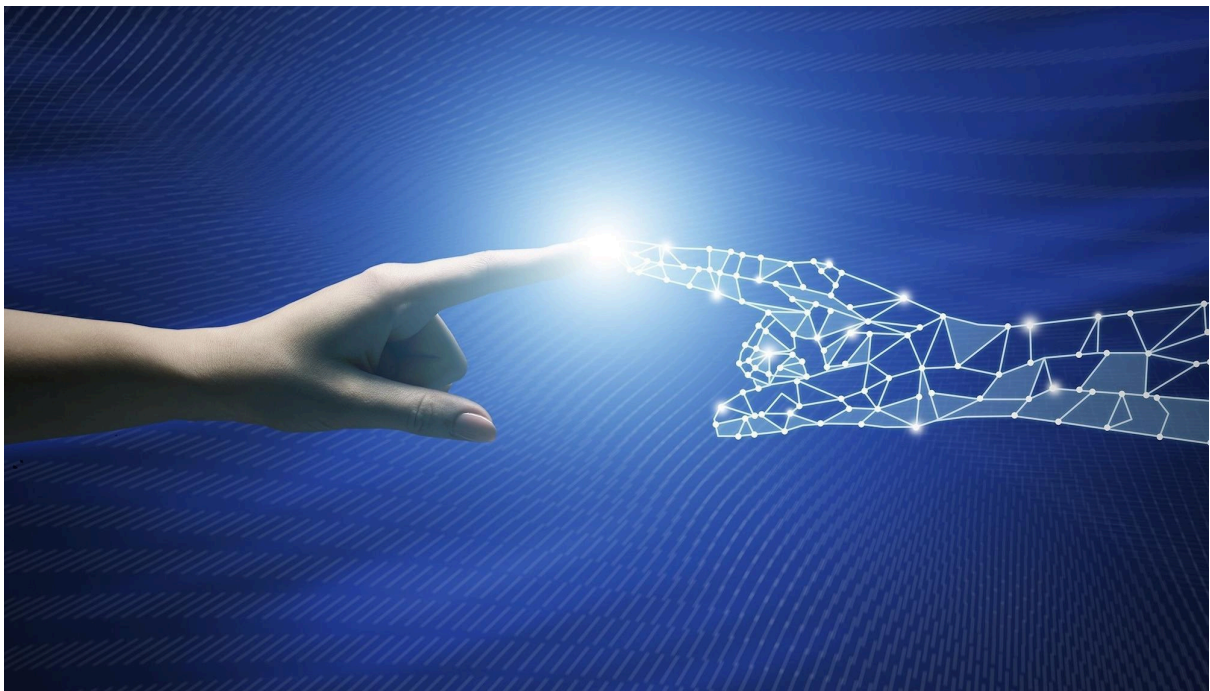
Project Overview

Data Classification is a machine learning technique used to categorize data into predefined classes. In this project, an AI model is trained using a dataset so that it can predict the category of new data.

For example:

- Email → Spam or Not Spam
- Flower → Iris Setosa, Iris Versicolor, Iris Virginica
- Student Result → Pass or Fail

The project demonstrates the complete supervised learning process including data collection, training, testing, and prediction.



Objective

The main objectives of this project are:

Step 1: Understand Data Classification

Learn how machines identify patterns and classify information.

Step 2: Load a Dataset

Import a dataset into Python for analysis.

Step 3: Prepare Data

Separate input features and output labels.

Step 4: Train the Model

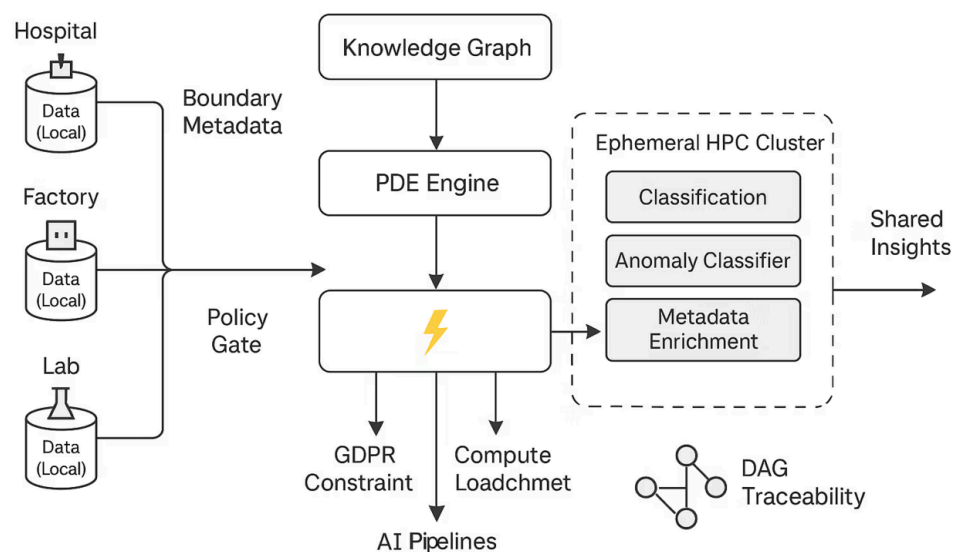
Teach the AI model using training data.

Step 5: Test the Model

Evaluate the model's performance using unseen data.

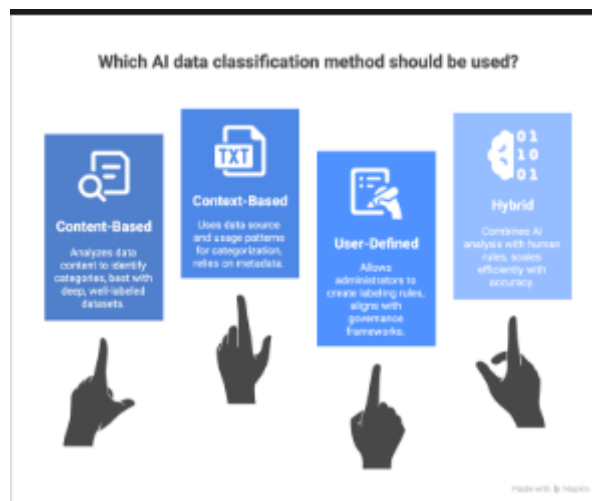
Step 6: Make Predictions

Predict the category of new data samples.



Problem Statement

Manually classifying large amounts of data is time-consuming and prone to errors. Organizations need intelligent systems that can automatically classify information quickly and accurately. The problem is solved by using Machine Learning classification algorithms that learn from historical data and predict future categories.



Software requirement

Software requirements are the programs, tools, and libraries needed to develop, train, and test the AI classification model. These software components help in data processing, machine learning, coding, and result analysis.

5.1 Python 3.x

What is Python?

Python is a high-level, easy-to-learn programming language widely used in Artificial Intelligence, Machine Learning, Data Science, and Web Development.

Why is Python Used?

- Simple and readable syntax.
- Large collection of AI and Machine Learning libraries.
- Easy to learn for beginners.
- Fast development and testing.

Role in This Project

Python is used to:

- Write the classification program.
- Load and process the dataset.
- Train the machine learning model.
- Predict output classes.

Jupyter Notebook

What is Jupyter Notebook?

Jupyter Notebook is an interactive development environment where Python code can be written and executed step by step.

Why is Jupyter Notebook Used?

- Easy code execution.
- Displays outputs immediately.
- Supports graphs and visualizations.
- Helpful for learning Machine Learning concepts.

Role in This Project

- Writing classification code.
- Testing machine learning models.
- Viewing outputs and accuracy results.

Features

- Interactive coding.
 - Automatic result display.
 - Supports Python and Data Science libraries.
-

5.3 Visual Studio Code (VS Code)

What is VS Code?

VS Code is a popular source code editor developed by Microsoft.

Why is VS Code Used?

- Lightweight and fast.
- Supports Python programming.
- Provides syntax highlighting.
- Easy debugging tools.

Role in This Project

- Writing and editing Python programs.
- Running machine learning scripts.
- Managing project files.

Features

- Auto-completion.
- Error detection.

- Extension support.
 - Integrated terminal.
-

5.4 NumPy Library

What is NumPy?

NumPy (Numerical Python) is a Python library used for numerical and mathematical operations.

Why is NumPy Used?

Machine Learning algorithms work with numbers and arrays. NumPy provides efficient handling of large datasets.

Role in This Project

- Store dataset values.
- Perform mathematical calculations.
- Process input features.

Features

- Fast computation.
- Array manipulation.
- Mathematical functions.

5.5 Pandas Library

What is Pandas?

Pandas is a Python library used for data analysis and data manipulation.

Why is Pandas Used?

Datasets are often stored in tables. Pandas helps organize, clean, and analyze data efficiently.

Role in This Project

- Load datasets.
- Read CSV files.
- Handle missing values.
- Display data in table format.

Features

- DataFrame support.
- Data filtering.
- Data cleaning.
- Statistical analysis.

Scikit-Learn Library

What is Scikit-Learn?

Scikit-Learn is one of the most popular Machine Learning libraries in Python.

Why is Scikit-Learn Used?

It provides ready-made Machine Learning algorithms that can be trained and tested easily.

Role in This Project

- Load the dataset.
- Split data into training and testing sets.
- Train the classification model.
- Measure prediction accuracy.

Theory

The theory section explains the basic concepts and working principles used in the Data Classification Using AI project. It helps us understand how Artificial Intelligence and Machine Learning are used to classify data into different categories.



Artificial Intelligence is a branch of computer science that enables machines to perform tasks that normally require human intelligence

Main Features of AI

1. Learning: AI systems learn from data and experience.
2. Reasoning: They can make decisions based on available information.
3. Problem Solving: AI can solve complex problems automatically.
4. Prediction: AI predicts outcomes using past data.

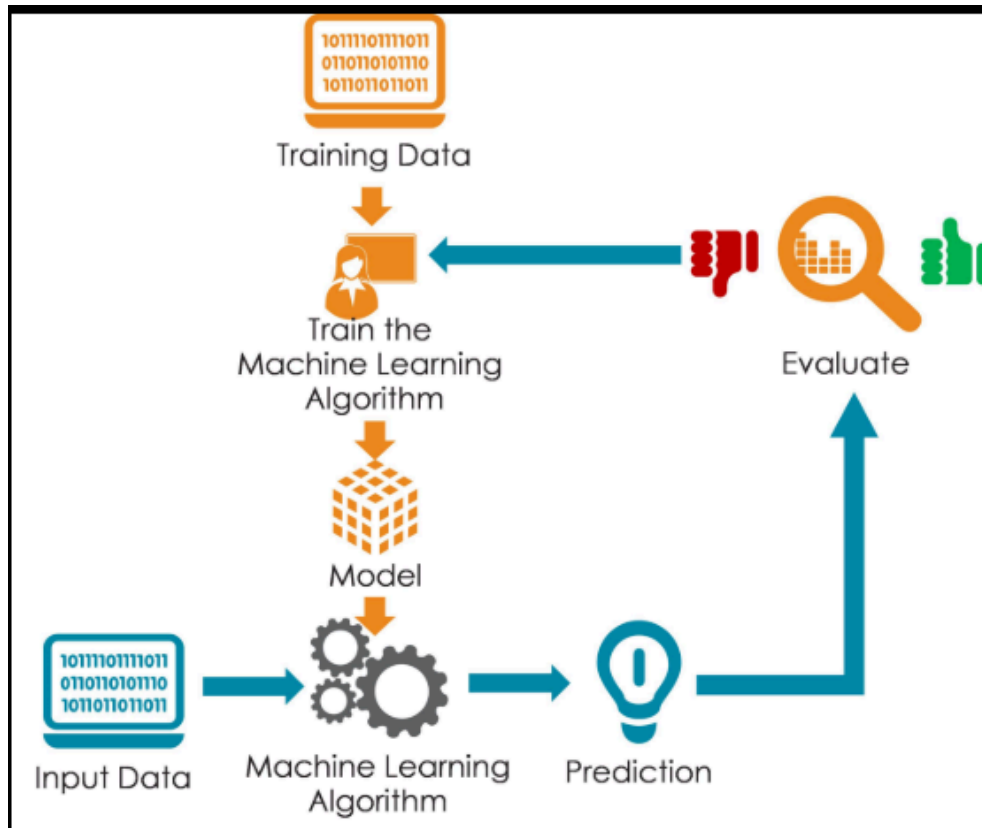
Examples of AI

- Voice assistants (Siri, Alexa, Google Assistant)
- Self-driving cars
- Recommendation systems (Netflix, YouTube)
- Chatbots and virtual assistants
- Face and speech recognition systems

Role in This Project

In this project, AI is used to build a system that can classify data automatically based on patterns learned from a dataset.

Machine Learning (ML)



Machine Learning is a subset of AI in which computers learn from data without being explicitly programmed.

How Machine Learning Works

1. Collect data.
2. Train a model using the data.
3. The model learns patterns from the data.
4. Use the trained model to make predictions on new data.

Types of Machine Learning

1. Supervised Learning

The model learns from labeled data (input + correct output).
Example: Predicting whether an email is spam or not spam.

2. Unsupervised Learning

The model finds patterns in unlabeled data. Example:
Grouping customers with similar behavior.

3. Reinforcement Learning

The model learns by interacting with an environment and receiving rewards or penalties.

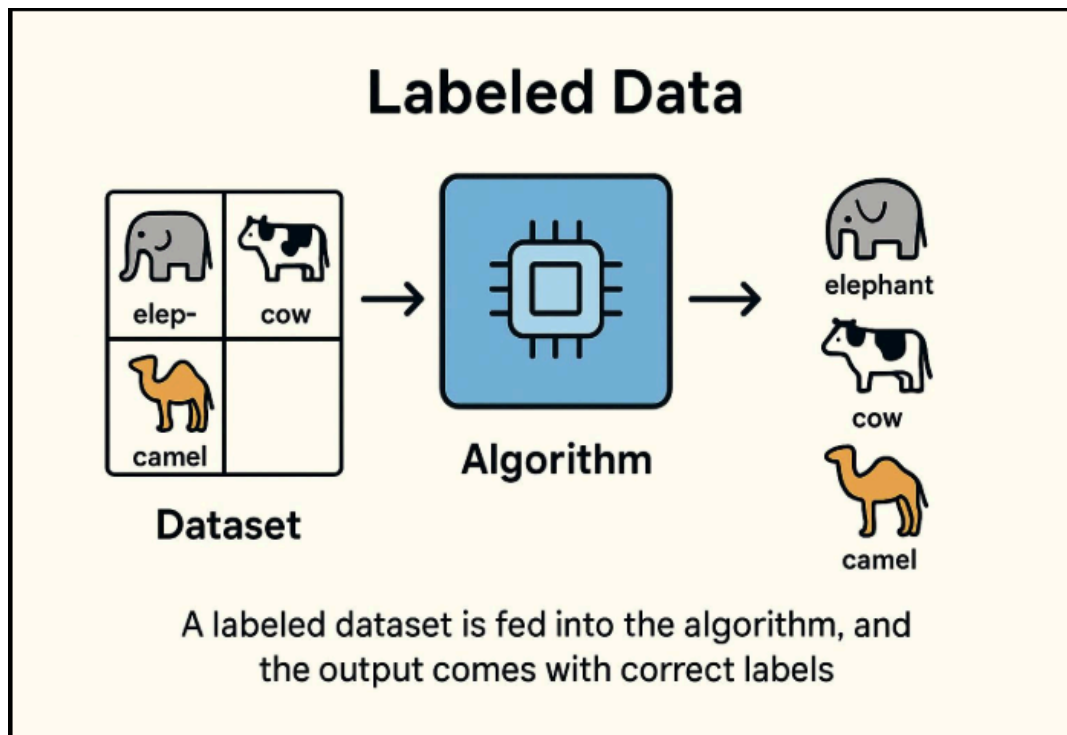
Role in This Project

This project uses Supervised Learning because the dataset already contains correct class labels.

Supervised Learning

Supervised Learning is a machine learning technique where the model is trained using labeled data. Each training example contains:

- Input Features (X): The data given to the model.
- Output Label (y): The correct category or answer.



Working Principle

1. The dataset contains input features and output labels.
2. The model learns the relationship between inputs and outputs.
3. After training, the model predicts the output for new unseen data.

Example

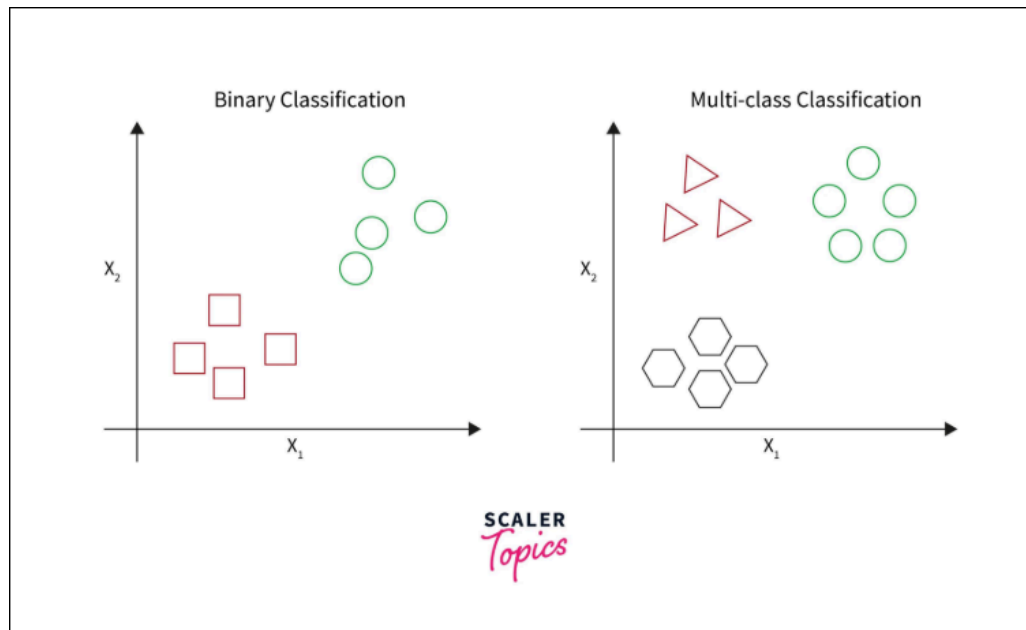
Input Features

**Output
Label**

Petal length, petal width, Flower type
etc.

Here, the model learns how flower measurements relate to flower categories.

Data Classification



Classification is a supervised learning task where data is assigned to predefined categories or classes.

Types of Classification

1. Binary Classification

Data is divided into two classes.

Example: Spam / Not Spam

2. Multi-Class Classification

Data is divided into more than two classes.

Example: Iris Setosa / Iris Versicolor / Iris Virginica

Role in This Project

This project performs multi-class classification using the Iris flower dataset.

Datasets used- Iris Datasets

The Iris dataset is one of the most famous datasets in machine learning. It contains measurements of iris flowers from three species.

Dataset Details

Feature	Description
Sepal Length	Length of the outer leaf of the flower.
Sepal Width	Width of the outer leaf of the flower.
Petal Length	Length of the inner colorful petal.
Petal Width	Width of the inner colorful petal

Target Classes

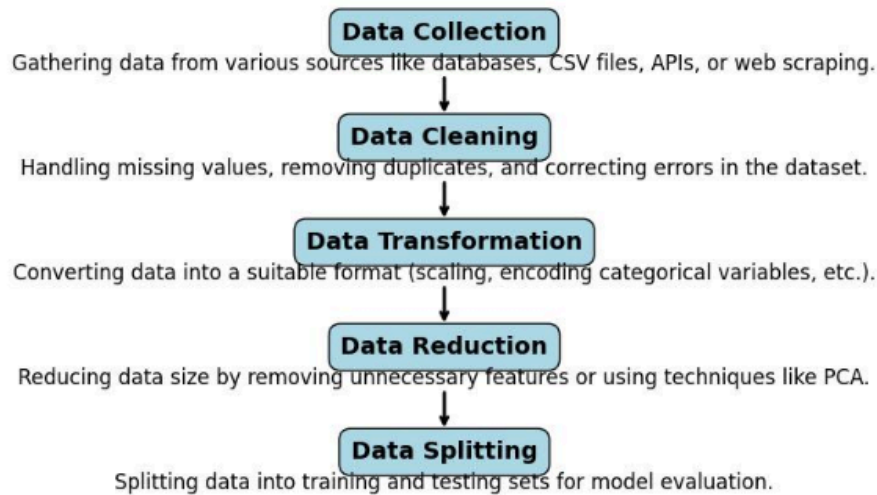
- Setosa
- Versicolor
- Virginica

Why Iris Dataset is Used

- Simple and easy to understand.
- Small dataset suitable for beginners.
- Commonly used for learning classification algorithms.

Data Preprocessing

Steps Involved in Data Preprocessing



Steps in Data Preprocessing

1. Loading the Dataset

The dataset is loaded into Python using Scikit-Learn.

2. Separating Features and Labels

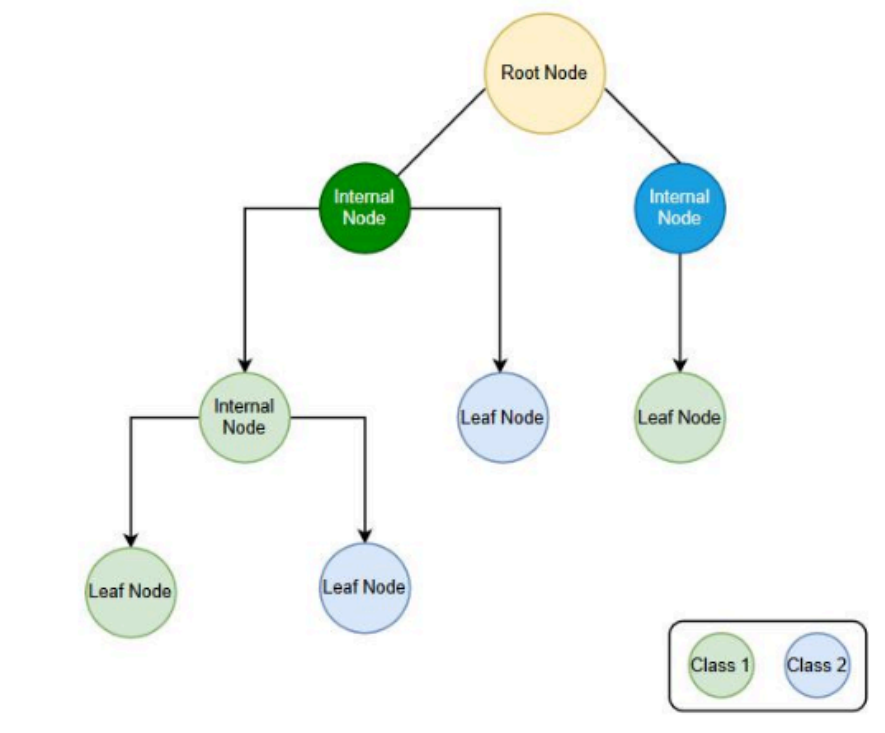
- X = Input features (flower measurements)
- y = Output labels (flower types)

3. Splitting the Dataset

The dataset is divided into:

- Training Data (80%) – used to train the model.
- Testing Data (20%) – used to evaluate the model.

Classification Algorithm Decision Tree



A Decision Tree is a supervised learning algorithm used for classification and prediction. It works like a flowchart where decisions are made based on feature values.

How Decision Tree Works

1. Start with the full dataset.
2. Select the best feature to split the data.
3. Create branches based on feature values.
4. Repeat the process until data is classified into classes.

Advantages of Decision Tree

- Easy to understand and visualize.
- Works well for classification problems.
- Requires less data preprocessing.
- Fast prediction speed.

Implementation in Python

Here:

- `DecisionTreeClassifier()` creates the model.
- `fit()` trains the model using training data.

Model Training

Training is the process where the model learns patterns from the training data.

Training Process

1. Input features (`X_train`) and labels (`y_train`) are given to the model.
2. The model analyzes relationships between features and classes.
3. It creates rules to classify future data.

Code

Result

The model becomes trained and ready to make predictions.

Model Testing and Accuracy

After training, the model is tested using unseen data to measure its performance.

Prediction

The model predicts classes for the testing data.

Accuracy Calculation

Accuracy shows the percentage of correct predictions.

Accuracy shows the percentage of correct predictions.

Formula of Accuracy

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

Algorithm

Step 1

Import required libraries.

Step 2

Load the Iris dataset.

Step 3

Separate features and labels.

Step 4

Split data into training and testing sets.

Step 5

Create classification model.

Step 6

Train model using training data.

Step 7

Test model accuracy.

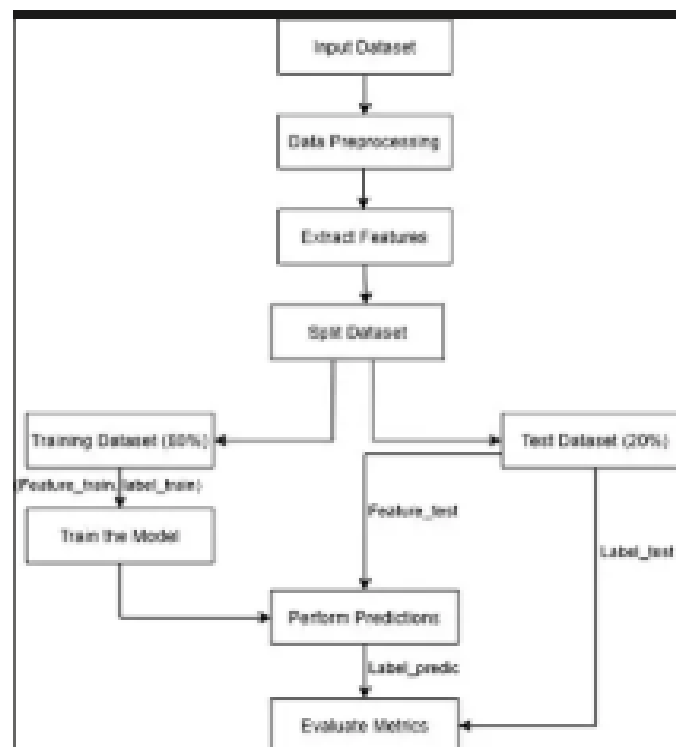
Step 8

Predict new data.

Step 9

Display results.

Flowchart



Program Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier()

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)
```

```
print("Predicted Class:", iris.target_names[prediction[0]])
```

Output

```
Accuracy: 1.0
```

```
Predicted Class: setosa
```

Advantages

1. Fast prediction.
2. High accuracy.
3. Reduces manual effort.
4. Easy to implement.
5. Useful in real-world applications.

Limitations

1. Depends on data quality.
2. May overfit small datasets.
3. Requires training data.
4. Accuracy decreases with poor data.

Future Enhancements

1. Use larger datasets.
2. Apply advanced algorithms.
3. Build a graphical interface.
4. Deploy as a web application.
5. Integrate real-time data.

Applications

- Email Spam Detection
- Medical Diagnosis
- Fraud Detection
- Face Recognition
- Customer Segmentation
- Product Recommendation Systems

Conclusion

This project demonstrates the implementation of a basic AI classification model using supervised learning. The model learns from training data and predicts the class of new data with high accuracy. The project helps understand the fundamental concepts of Artificial Intelligence, Machine Learning, and Data Classification.

THANK YOU